

Document Summary Assistant – Approach

The **Document Summary Assistant** is a full-stack web application designed to automatically extract, summarize, and improve the readability of documents. The application uses **Node.js and Express** for backend processing and **HTML, CSS, and JavaScript** for the frontend.

Users can upload PDF documents or scanned images through a simple drag-and-drop interface. For PDFs, text is extracted using `pdf-parse`, while scanned images are processed using `Tesseract.js` for OCR. The extracted content is then processed using a built-in Natural Language Processing approach that identifies important keywords and sentences.

Based on the extracted information, the application generates **Short, Medium, and Long summaries**, along with key takeaways in bullet-point format. It also calculates a readability score and provides writing improvement suggestions, including reducing long sentences, removing unnecessary words, and improving document structure with clear headings.

A key focus of the project was keeping the application **self-contained and cost-effective**. It does not depend on external AI APIs, API keys, or paid services. This makes the solution easy to deploy, maintain, and use while demonstrating practical document-processing and NLP concepts.

Application: <https://document-summary-assistant-silk.vercel.app>

GitHub: <https://github.com/MokshithReddy2/document-summary-assistant/>